

SPIDER 2.0: Driver Distraction and Visual Attention

David L. Strayer and Amy S. McDonnell

Department of Psychology, The University of Utah, Salt Lake City, Utah, USA;
email: david.strayer@utah.edu

Annu. Rev. Vis. Sci. 2025. 11:521–40

First published as a Review in Advance on
April 11, 2025

The *Annual Review of Vision Science* is online at
vision.annualreviews.org

<https://doi.org/10.1146/annurev-vision-110423-025626>

Copyright © 2025 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.



**ANNUAL
REVIEWS** **CONNECT**

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

Keywords

cell phones, crashes, driving, distraction, divided attention, multitasking, vision, inattentional blindness, situational awareness

Abstract

Driving is a complex multisensory experience that requires the integration of various sensory inputs to maintain effective situational awareness, with vision and visual attention being paramount for safe driving. However, multitasking significantly degrades a driver's situational awareness and causes them to overlook or misjudge important aspects of their environment, such as pedestrians, road signs, or other vehicles. It also impairs a driver's ability to visually scan for hazards and process vital information, reducing their capacity to notice and respond to changes on the roadway. Multitasking can also induce inattentional blindness, causing drivers to miss important information directly in their line of sight. Beyond diminished visual attention, multitasking also slows reaction times to detected events, increasing the likelihood and severity of crashes. This article discusses the central role that visual attention plays in a driver's situational awareness, examines common methods for assessing visual attention while driving, and presents an updated review of the SPIDER (scanning, predicting, identification, decision-making, and executing a response) model of driver awareness with a focus on visual distraction.

INTRODUCTION

Driving involves a complex set of learned behaviors that enable safe navigation of a motor vehicle from one location to another. This task demands good situational awareness, allowing for motorists to recognize actual and potential hazards on the roadway (Strayer & Fisher 2016). Drivers must execute timely evasive actions, such as steering or braking, should a threat materialize. Additionally, maintaining lateral and longitudinal control is essential; drivers should remain within their lane of travel and keep a safe following distance from other vehicles. Drivers must also adjust their speed according to driving conditions, which may require traveling below the posted speed limit. Safe driving involves not only effective situational awareness and timely evasive actions but also managing distractions, predicting the behaviors of other road users, and adapting to adverse weather conditions. Additionally, drivers must be prepared for emergencies, understand vehicle dynamics, and communicate effectively with others on the road while adhering to traffic laws.

Thus, driving is a skill that can take years to master. In fact, motor vehicle crashes are the leading cause of accidental injury deaths in the United States (NHTSA 2015). Inexperienced drivers face an even greater risk of being involved in such crashes (NHTSA 2023). The introduction of graduated driver licensing laws, which impose restrictions on novice drivers—like prohibiting nonfamily members in the vehicle for the first 6 months—has led to significant reductions in accidents, with fatal crashes decreasing by 38% and injury crashes by 40% (Baker et al. 2007, Hedlund et al. 2003). Crash rates tend to plateau around the age of 25, coinciding with the full maturation of the prefrontal cortex (Watson et al. 2011). This suggests that attentional control and decision-making abilities improve as the brain matures, reducing risky behaviors and poor judgment that contribute to crashes by younger drivers (Foy et al. 2016).

Driving is a complex multisensory experience that demands the integration of various sensory inputs to maintain effective situational awareness. Of the five human senses, vision is the most critical for driving. Drivers rely on visual input to maintain lane position, sustain an appropriate following distance from the vehicle ahead of them, and respond to changes in driving conditions. Vision enables the detection of other vehicles, pedestrians, traffic signals, and potential hazards in the environment, as well as the assessment of speed and depth perception. Fixating on objects in central vision is crucial for driving tasks such as reading road signs, gauging the speed and distance of nearby vehicles, and navigating lane markers. Furthermore, drivers typically steer in the direction they are looking and gaze in the direction they are steering (Cooper et al. 2013, Land & Lee 1994, Navarro et al. 2020). In addition to forward vision, motorists use side and rearview mirrors to maintain situational awareness of surrounding traffic (Large et al. 2016, Mazzae et al. 2023, Mole & Wilkie 2017). Thus, for effective driving, drivers must continuously scan their environments—not only to effectively direct the vehicle to where they are headed but also to be ready to respond to unexpected events.

Foveal vision is not the only aspect of vision that is critical for driving. Peripheral vision plays an equally important role, as it enables drivers to detect motion and identify objects outside their central line of sight, enhancing overall situational awareness (Gugerty 2011, Wolfe et al. 2017). Other characteristics of vision like night vision and color discrimination assist with recognizing brake lights, turn signals, and road signs in varying environmental lighting conditions (Jones et al. 2022, Wood 2020). Any impairment in visual acuity, contrast sensitivity, or field of vision can significantly increase the risk of accidents (Anderson et al. 2021).

While vision is central to safe driving, auditory and haptic senses further enhance a driver's ability to process and respond to dynamic conditions (Durso et al. 2007; Endsley 1995, 2015; Geitner et al. 2019; Horrey et al. 2006; Kass et al. 2007). Auditory cues provide drivers with critical information that may not be visible, such as the sound of a horn alerting to a potential collision,

a car engine indicating proximity to another vehicle, or warning beeps from advanced driver-assistance systems detecting nearby obstacles (Ho & Spence 2005, Ju et al. 2020, Wang et al. 2020). Similarly, haptic feedback—such as the vibrations felt through the steering wheel, seat, or even the vehicle’s pedals—acts as a nonvisual, nonauditory form of communication (Di Campli San Vito et al. 2019, Xin et al. 2021). Subtle vibrations from road surface changes, such as rumble strips or uneven terrain, offer immediate tactile warnings that alert the driver to adjust their course. For instance, lane departure systems often use steering wheel vibrations to notify drivers if they begin to drift, while antilock braking systems deliver pedal feedback to indicate loss of traction during sudden stops. These haptic cues help maintain situational awareness, especially in scenarios where visual or auditory distractions might otherwise impair decision-making (for a review of haptic technologies in the vehicle, see Gaffary & Lécuyer 2018).

MULTITASKING

Multitasking diverts a driver’s attention from the primary task of driving, significantly increasing the risk of a crash (Gershon et al. 2019, McEvoy et al. 2005, Redelmeier & Tibshirani 1997).¹ A common form of multitasking involves the use of a cell phone to talk, text, or interact with a device while driving (National Center for Statistics and Analysis 2021). Driving and cell phone use are functionally independent tasks (Strayer et al. 2022a,b), each with separate goals—transportation versus communication—and distinct stimuli, such as the driving environment versus the content of the conversation, as well as different responses (e.g., steering and braking versus talking and listening). Regan et al. (2011, p. 1772) defined this type of multitasking as “the diversion of attention away from activities critical for safe driving toward a competing activity, which may result in insufficient or no attention to activities critical for safe driving.”

In contrast, interdependent tasks work together toward the same overall goal, such as maintaining vehicle control by checking the speedometer, adjusting the accelerator to manage speed, and steering to stay in the lane. Regan et al. (2011, p. 1775) defined the impairment from performing interdependent tasks as “insufficient or no attention to activities critical for safe driving brought about by the driver focusing attention on one aspect of driving to the exclusion of another, which is more critical for safe driving.” Therefore while interdependent tasks may complement one another to achieve the shared goal of maintaining control of the vehicle, misallocating attention among them can also compromise safety.

Driving impairments can occur due to competing demands on visual processing, such as when drivers divert their gaze from the road to check a cell phone display. This diverts crucial visual attention away from detecting hazards, traffic signals, or other vehicles (Amini et al. 2023). Additionally, manual distractions arise when drivers remove their hands from the steering wheel to perform manual tasks such as texting, eating, or adjusting the radio, reducing their ability to control the vehicle (Moslemi et al. 2021). Last, cognitive distractions arise when a driver’s focus shifts away from processing the essential information needed for safe driving, such as when thinking about non-driving-related tasks or engaging in conversation, leading to slower reaction times and decreased situational awareness (D’Addario & Donmez 2019, Salvucci 2019). Together, these distractions compromise overall driving performance and significantly increase the likelihood of accidents (NSC 2024).

¹In this article, the terms multitask and dual task are used interchangeably. A dual-task situation involves participants performing two functionally independent tasks simultaneously. Multitasking, on the other hand, is a broader term that refers to performing two or more independent tasks either concurrently or by alternating between them (for a review of different forms of multitasking, see Koch et al. 2018).

Visual and manual sources of interference often co-occur and are examples of structural interference (Strayer et al. 2019; Wickens 1980, 2008). A typical example is using a cell phone to send or read a text message, where the driver cannot simultaneously focus on both the phone's display and the road ahead. In this scenario, both visual (gazing at the phone) and manual (holding the phone and typing) demands compound, amplifying distraction and increasing the likelihood of impaired driving performance. Naturalistic driving studies have shown that glancing away from the road significantly increases crash risk, with longer glances correlating with greater crash likelihood (Simons-Morton et al. 2014). Atwood et al. (2018) found that drivers who text more frequently face a higher risk of crashing, with the rate of severe crashes increasing by 0.58% for every additional text sent per day. Simply put, failing to look at the road leads to crashes or veering off the roadway.

When humans perform two or more attention-demanding tasks concurrently, one task often prospers at the expense of the other(s) (Kahneman 1973; Navon & Gopher 1979; Norman & Bobrow 1975; Strayer & Drews 2007a; Wickens 1980, 1984, 2008). For instance, during a cell phone conversation, a sawtooth pattern of resource allocation emerges, with driving and the conversation competing for the same limited cognitive resources (Castro et al. 2019, Drews et al. 2008, Tillman et al. 2017). Notably, drivers experience greater interference from speaking than from listening during phone conversations (Strayer et al. 2017). Furthermore, the cognitive distraction caused by cell phone conversations is similar whether using handheld or hands-free devices (Caird et al. 2018, McEvoy et al. 2005, Redelmeier & Tibshirani 1997, Strayer & Johnston 2001, Strayer et al. 2015).

Multitasking can also create cross talk between tasks, particularly when they involve similar or overlapping information (e.g., Bergen et al. 2013; Navon & Miller 1987; Pashler 1994a,b). Notably, Howard et al. (2020) used evidence accumulation modeling to demonstrate that multitasking differs from simply increasing task difficulty. While both reduce the rate of evidence accumulation, multitasking additionally raises the response threshold, whereas increased task difficulty does not. This finding indicates that multitasking imposes an extra cognitive cost, reflecting a strategic adjustment made to accommodate performing multiple tasks simultaneously (Howard et al. 2020).

MEASURING VISUAL ATTENTION AT THE WHEEL

Given the critical role that vision and visual attention play in maintaining situational awareness while driving, it is essential to explore how visual attention and the visual distraction arising from multitasking can be effectively assessed in real time. This is an essential step in identifying the conditions in which visual attention may deteriorate and in developing reliable driver monitoring systems that track visual engagement and can alert the driver to reengage attention back to the roadway (Biondi et al. 2024a).

Visual occlusion is a method used in driving research to simulate moments when visual information is temporarily unavailable, mimicking real-world situations where a driver's view may be obstructed or attention is momentarily diverted (Senders et al. 1967). This technique involves deliberately blocking or occluding part of a driver's visual field for short durations to assess their ability to anticipate and react to changing traffic conditions when deprived of constant visual input (e.g., Kujala & Sarkar 2024). This can be done by turning a display on and off or by physically blocking a driver's vision with a screen or eye goggles (Foley 2008). By replicating these real-world scenarios in which visual input is momentarily lost—such as when drivers glance away from the roadway to check a phone or dashboard display—researchers can examine how well drivers compensate for the loss of visual information, estimate vehicle control, and predict potential hazards when their view is blocked (Gilroy et al. 2019). This examination allows for insights into how long drivers can safely manage their vehicle without updated visual cues (Foley 2008). The time

between occlusions is critical; if drivers can maintain control and make decisions during occlusion periods, it suggests they have effectively processed and retained necessary visual information. The technique also reveals how dependent drivers are on continuous visual input and how quickly they can adapt to changes in visibility. For safety reasons, visual occlusion methods are often used in driving simulator or closed road studies and must be coupled with an assessment of successful driving performance (Kujala et al. 2023). In general, the technique provides insights into the minimum safe eyes-off-road time for drivers and the attentional demand and spare visual capacity of specific tasks (for a review, see Kujala et al. 2023).

Visual attention can also be assessed with simple stimulus response tasks in which drivers respond when they detect a randomly presented visual stimulus in their forward line of sight. For example, the detection response task (DRT) (ISO 2016) is a standardized protocol designed to measure the cognitive demands associated with secondary-task engagement involving visual-manual, voice-based, or haptic interfaces. In this task, participants are presented with a simple stimulus (e.g., a light or vibrotactile device) every 3–5 s, and they must press a button when they detect the stimulus. The DRT is highly sensitive to variations in cognitive load; as mental workload increases, reaction times to the DRT stimulus lengthen, and the likelihood of detecting the stimulus (i.e., the hit rate) decreases (e.g., Biondi et al. 2024b, Castro et al. 2019). The LED, light stimulus variant of the DRT provides a measure of visual attention (ISO 2016). This variation of the DRT involves an LED positioned within the driver's forward field of view, and participants are instructed to press a finger switch whenever they detect the LED illuminating. Miss rates in response to the LED provide a brief, accessible measure of visual attention, such that if the driver is not looking at the forward roadway, they will not see the intermittently illuminated LED and thus will miss the appropriate button response. Several studies have validated the method for its sensitivity to visual task demands (e.g., Bruyas & Dumont 2013, Castro et al. 2016, Ranney et al. 2014, van Winsum 2018) and changes in cognitive workload (Castro et al. 2016; ISO 2016; McDonnell et al. 2021a,b; Strayer et al. 2018). However, while effective, this approach does not capture eyes-off-road time and is less precise than eye tracking or glance coding, due to the intermittent onset of the LED stimulus (typically every 3–5 s), which might not always align with moments when drivers glance away from the road ahead.

Eye tracking can offer insights into how individuals process visual scenes and allocate visual attention with high temporal resolution. Recording eye movements has been a key method for understanding the visual nature of driving tasks (for reviews, see Crundall & Underwood 2011, Grüner & Ansorge 2017, Kapitaniak et al. 2015). Driver-facing cameras offer a cost-effective and portable solution, providing valuable insights into where and how drivers allocate their visual attention in real time.

Fixation rate can be used to measure how long and frequently drivers focus on specific areas, such as road signs or other vehicles, providing insights into which roadway regions and objects are prioritized by drivers at any given moment. Saccades, or the rapid eye movements between fixations, can be analyzed to assess how efficiently drivers scan their environment, indicating their ability to quickly shift focus and detect potential hazards. Visual attention can also be assessed by coding eye glances on and off the forward roadway. Off-road glances can be categorized as either task relevant, such as looking at side mirrors, or task irrelevant, such as looking at a passenger, cell phone, or the vehicle's touchscreen (e.g., Biondi & Jajo 2024). Blink rate and duration have also been used to detect driver workload (e.g., Benedetto et al. 2011, Yahoodik et al. 2021) and drowsiness (Cooper et al. 2023, Shamini et al. 2022), both of which inevitably interact with visual attention.

Gaze dispersion/concentration is also used to understand driver visual attention (Gonçalves et al. 2020, Louw & Merat 2017, Wang et al. 2014). Gaze dispersion refers to the distribution of

a person's visual attention across different areas of their visual field. It quantifies how widely or narrowly a person's gaze is spread while observing their environment. In the context of driving, gaze dispersion indicates how drivers allocate their visual focus among various elements, such as the road, other vehicles, traffic signs, and potential hazards. A wider gaze dispersion may suggest divided attention, whereas a more concentrated gaze can indicate focused attention on relevant driving tasks (Wang et al. 2014).

Electroencephalography (EEG) can also be used to measure a driver's visual engagement with their surroundings in real time (Lohani et al. 2019). EEG records brain activity via electrodes positioned on the scalp, allowing for the direct recording of neural responses to environmental demands with millisecond precision. Notably, EEG is noninvasive and can be made portable, making it a practical tool for use in both driving simulators and on-road assessments. The oscillatory activity that makes up the EEG signal can be decomposed into various frequency bands [e.g., theta ($\sim 4\text{--}8$ Hz), alpha ($\sim 8\text{--}12$ Hz), beta ($\sim 12\text{--}30$ Hz)] associated with different neurocognitive functions (Lohani et al. 2019).

Alpha oscillations (8–12 Hz) distributed across the visual cortex (posterior regions of the scalp) can be used to assess visual attention (Peylo et al. 2021, Romei et al. 2008, Sauseng et al. 2005, Thut et al. 2006). Alpha oscillations are inversely related to visual attention, such that when visual attention is high, alpha power is low, and when visual attention is low, alpha power is high (Foxe & Snyder 2011, Toscani et al. 2010). Posterior alpha oscillations were among the earliest brain rhythms identified in EEG research largely because of their robust response to visual engagement; when the eyes are closed, alpha oscillations surge dramatically, highlighting their sensitivity to changes in visual input (Adrian & Matthews 1934, Berger 1929, Goldman et al. 2002). Posterior alpha activity has been used in driving contexts as a biomarker for driver visual engagement with their environment (e.g., Brooks et al. 2018; McDonnell et al. 2021a,b, 2023). For example, McDonnell and colleagues (2021a,b) recorded EEG activity while individuals drove on two unique roadway conditions—a straight interstate and a winding, mountainous road. Drivers exhibited significantly lower posterior alpha power on the curvy, mountainous road, indicating heightened visual engagement in the more challenging driving demands associated with maintaining lane position while navigating steep curves. Such work underscores the utility of posterior alpha oscillations as a reliable biomarker for assessing visual attention in dynamic, real-world contexts. By tracking changes in alpha power, researchers can gain valuable insights into how drivers respond to varying levels of visual and cognitive demand on the road.

SPIDER 2.0

Our brains can process only a finite amount of incoming information at any given time, making it difficult to attend to every detail in a dynamic environment (Kahneman 1973). As discussed, driving places exceptionally high demands on all our sensory systems, particularly vision, as drivers must constantly filter and prioritize visual input while processing auditory, tactile, and spatial information to make split-second decisions. Driver distraction arises from limited attentional capacity (Strayer et al. 2022a,b). Safely operating a motor vehicle requires several mental processes, including visually scanning the driving environment, predicting potential hazards—even those that are not immediately visible—identifying objects within the line of sight, deciding whether an action is necessary, and executing timely responses. Collectively, these processes are referred to as SPIDER (Strayer & Fisher 2016).

SPIDER is essential for establishing and maintaining situational awareness and facilitates expectancy-based processing of the driving scene. **Figure 1** illustrates the relationship between limited attentional capacity, SPIDER-related cognitive processes vital for safe driving, and

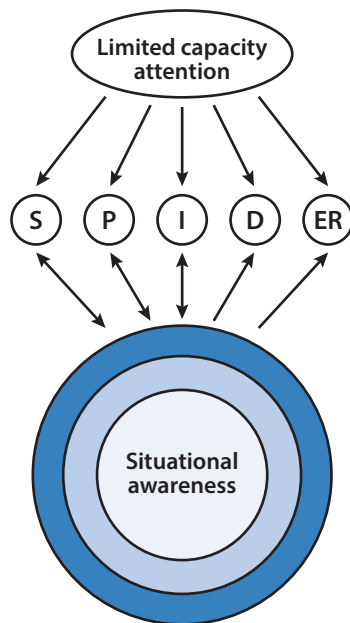


Figure 1

The SPIDER model of driver distraction is an acronym that stands for scanning, predicting, identification, decision-making, and executing a response. These SPIDER-related processes rely on our limited attentional capacity and are essential for maintaining situational awareness at the wheel. When a driver is distracted, their situational awareness diminishes, as depicted in the figure by progressively smaller and lighter-colored circles. The bidirectional arrows illustrate how situational awareness is informed and updated by the SPIDER processes, facilitating expectancy-based processing of the driving environment. The degradation of situational awareness impairs driving performance and increases the risk of a crash.

situational awareness. When motorists engage in non-driving-related activities—those functionally independent of the driving task—attention is diverted from driving, leading to degraded situational awareness and an increased risk of crashes. In the following paragraphs, we present an updated review of the literature regarding SPIDER-related processes and driver distraction.

Visual Scanning

Visually scanning the forward roadway, peripheral areas, rear and side mirrors, and vehicle blind spots is essential for safe driving. This process aligns with Endsley's level 1 situational awareness, which involves perceiving the critical elements in the current environment that may affect driving performance (Endsley 1995, 2015). Engaging in a secondary task has been shown to decrease visual scanning (Horrey et al. 2006; Mackenzie & Harris 2017; McCarley et al. 2001, 2004; Victor et al. 2005). Moreover, as visual scanning diminishes, drivers may develop tunnel vision, focusing only on a narrow segment of the roadway, further hindering their situational awareness and ability to anticipate potential threats (van Winsum 2019).

Even secondary tasks that do not have an explicit visual component can lead to drivers focusing their gaze on the center of the driving scene (Briggs et al. 2017, Engström et al. 2005, Goh et al. 2021, Harbluk et al. 2007, He et al. 2011, Recarte & Nunes 2000, Reimer 2009, Reimer et al. 2012, Strayer et al. 2017, Tsai et al. 2007, Victor et al. 2005). This concentration of gaze may also affect lateral lane position variability (Readeringer et al. 2002, Rogers et al. 2005, Wilson et al. 2008),

although some studies suggest differing outcomes (Cooper et al. 2013, Medeiros-Ward et al. 2014). Often, threats to safe driving first appear in the periphery, such as a distracted pedestrian crossing the roadway. The tunnel vision induced by multitasking reduces the likelihood of spotting these threats in time to take evasive action (van Winsum 2019).

Hazard Prediction

The dual mechanisms of control framework developed by Braver and colleagues (e.g., Braver 2012; Braver et al. 2008, 2009) identifies two modes of cognitive control: reactive and proactive. Reactive control is a bottom-up process that functions as a late-correction strategy in response to unexpected events, such as when a pedestrian suddenly steps out in front of a parked vehicle. In contrast, proactive control is a top-down process that operates anticipatorily to guide driving behavior. Hazard prediction, a critical component of safe driving, exemplifies proactive control and aligns with Endsley's level 3 situational awareness, which focuses on predicting future states of the environment (Endsley 1995, 2015).

Anticipatory glances toward specific areas of the roadway indicate that a driver is making top-down predictions about potential hazards in those locations. For instance, a driver may glance at an alleyway as they approach to ensure that no vehicles will pull out in front of them. Fisher (2015) studied drivers' eye movements at intersections and found that they often make an initial glance to the left and right as they approach and then a secondary glance just before entering the intersection. Multitasking has been shown to reduce these anticipatory glances toward potential hazards (Biondi et al. 2015; Ebadi et al. 2019; T. Taylor, L. Roman, K. McFeaters, M. Romoser, A. Borowsky, et al., unpublished data). Biondi et al. (2015) demonstrated that the likelihood of glancing for pedestrians in crosswalks diminished when drivers were engaged in a cognitively demanding secondary task. Beyond hazard prediction at intersections, Ebadi et al. (2019) replicated this pattern at work zones and marked midblock crosswalks. Similarly, T. Taylor, L. Roman, K. McFeaters, M. Romoser, A. Borowsky, et al. (unpublished data) found that secondary-task engagement caused experienced drivers to behave more like inexperienced drivers. This claim is further supported by Wright et al. (2016), who found that experienced drivers look more frequently at potential hazard locations than younger drivers do. Additionally, multitasking can induce a form of contextual blindness, characterized by impairments in the ability to use environmental cues to guide attention in the driving environment (Cooper & Strayer 2025). This contextual blindness hinders both the acquisition of new sequences and the execution of previously learned tasks.

Identification

People can look at their surroundings but still fail to see important information, resulting in what is known as look but fail to see (LBFTS) errors (Wolfe et al. 2022). These errors arise from the limited capacity of attention; observers must actively focus on what they are looking at or risk being blind to it (Mack & Rock 1998, Simons 2000, Simons & Chabris 1999). A classic study by Simons & Chabris (1999) illustrates this phenomenon: Participants tasked with counting the number of passes made by a basketball team failed to notice a person in a gorilla suit walking through the scene. The gorilla stopped, faced the camera, thumped its chest, and then exited, remaining visible for over 5 s. Remarkably, about half of the participants did not see it, demonstrating inattentional blindness (LBFTS). These errors are prevalent in driving situations (Herslund & Jorgensen 2002) and become more frequent when individuals are multitasking (Saryazdi et al. 2019, Strayer & Cooper 2015).

Drivers must attend to visual input to identify objects in their environment, enabling them to respond appropriately and promptly. Identification, or seeing, is crucial for safe driving and relates to Endsley's level 2 situational awareness, which involves comprehension of the current

situation (Endsley 1995, 2015). However, drivers engaged in secondary tasks often experience inattention blindness, failing to notice items within their field of view (Pizzighello & Bressan 2008, Saryazdi et al. 2019, Strayer & Drews 2007b, Strayer & Johnston 2001, Veitch & Alsos 2023). For a review on the influence of perceptual and cognitive load on inattention blindness, see Matias et al. (2022).

Strayer et al. (2003) examined drivers' recognition memory for objects they looked at while driving (as verified by eye tracking) and found that those talking on a cell phone exhibited a 50% reduction in recognition memory compared to baseline performance. In subsequent studies, these researchers varied the safety relevance of objects in the driving scene (e.g., pedestrians, other vehicles, parked cars, billboards) to assess whether more-safety-relevant items were less susceptible to inattention blindness. Unfortunately, they found no correlation between safety relevance and vulnerability and inattention blindness (Strayer et al. 2004).

While multitasking, drivers fail to strategically prioritize the processing of safety-critical information in the driving environment over the conversation on their cell phones. Supporting evidence for inattention blindness comes from event-related potential (ERP) studies using EEG. ERPs are deflections in the EEG waveform that occur in response to a discrete event in the driving environment. The amplitude of the P300 indexes the amount of attention allocated to a stimulus—the higher the amplitude is, the higher the attention allocated (Sirevaag et al. 1989). Research has shown that the amplitude of the P300 component—elicited by discrete driving-relevant events—decreases as secondary task demands increase (e.g., Leynes et al. 2018; Strayer & Drews 2007b; Strayer et al. 2004, 2015). This ERP pattern indicates that engagement in secondary tasks compromises the encoding and registration of traffic-related information.

Posterior alpha oscillations in the brain may also serve as an index of inattention blindness. As discussed above, alpha oscillations distributed across the visual cortex (posterior regions of the scalp) can be used to assess visual attention at the wheel (e.g., McDonnell et al. 2021b, 2023). Given that increased power in posterior alpha oscillations is associated with decreased visual attention (Paylo et al. 2021) and that a driver's ability to process peripheral visual information is diminished during secondary-task engagement, posterior alpha power is a promising metric for indicating momentary threats to inattention blindness. The use of posterior alpha power as an indicator of inattention blindness has been established by researchers in contexts beyond driving (Hutchinson et al. 2020, 2021), suggesting its promising applicability to driving scenarios. By monitoring posterior alpha activity, researchers can gain insights into the driver's attentional state and identify moments when critical visual information may be overlooked, enhancing our understanding of the neural mechanisms underlying inattention blindness in real-world driving scenarios.

Decision-Making

Drivers must assess multiple sources of information to make safe decisions when changing lanes, merging into traffic, or making left turns across oncoming vehicles. However, the ability to make sound judgments declines with age (Strayer 2015). For instance, multivehicle crashes at intersections account for 20–25% of fatal accidents among drivers under the age of 60. In contrast, the crash rate steadily increases after age 60, reaching 45% of fatal crashes by age 85. Intersections pose a significant mental challenge, requiring drivers to focus on traffic signals, cross traffic, pedestrians, and other vulnerable road users (NSC 2018). Meanwhile, the rates of single-vehicle crashes and multivehicle crashes outside intersections continue to decline with age. Older adults face heightened risks, especially when tasked with processing multiple sources of information, particularly during left turns at intersections (Rolison et al. 2018, Sifrit et al. 2011).

When drivers make unprotected left turns at intersections or across traffic, they must identify a safe gap in the flow of vehicles to complete the maneuver. Although dedicated left-turn signals

simplify this task, many intersections require drivers to assess what constitutes an acceptable gap based on the time needed to cross safely. Research indicates that older drivers experience a higher incidence of at-fault crashes during these gap-acceptance maneuvers (Evans 2004, Farmer et al. 1997, Staplin & Lyles 1991).

Moreover, multitasking can impair drivers' ability to accurately evaluate critical information necessary for safe decision-making (Choudhary & Velaga 2019). For instance, Cooper et al. (2009) found that drivers using hands-free cell phones were more likely to make unsafe lane changes. Similarly, Cooper & Zheng (2002) observed that distracted drivers often misjudged the size of gaps and the speed of oncoming vehicles, a trend that was particularly pronounced on wet roadways. Haque et al. (2016) further demonstrated that drivers distracted by hands-free cell phones maintained a narrower safety margin compared to their nondistracted counterparts.

The lane change task (ISO 2010, Zheng et al. 2023) is a standardized protocol designed to assess the impact of secondary tasks on driver decision-making. In this protocol, participants drive in the middle lane of a three-lane highway, where traffic in adjacent lanes is moving at varying speeds. When a decelerating vehicle is detected ahead—as done by collision warning systems found in modern vehicles—drivers must quickly assess the situation, factoring in the speed and position of other vehicles, and decide whether to safely switch to the right or left lane to avoid a potential collision (Zheng et al. 2023). This is an ecologically valid task that has been extensively used to explore driver decision-making capabilities under various driving conditions (e.g., Gonçalves et al. 2022, Wang et al. 2018). It draws on both foveal and peripheral visual processing, testing driver visual attention during complex decision-making tasks.

Execution of a Response

One of the key indicators of divided attention is a slowing of reaction time (Posner & Boies 1971, Posner & Keele 1967). When a driving situation necessitates an evasive action—like swerving to avoid a child running into the street or braking to prevent a collision with an oncoming vehicle—these reactions are often delayed (Atchley et al. 2017; Caird et al. 2008, 2018; Haque & Washington 2014; Horrey et al. 2006; Ishigami & Klein 2009). The impact of secondary tasks becomes more pronounced as the perceptual demands of the driving environment increase, such as when traffic volume rises (Strayer et al. 2003). Additionally, engaging in a secondary task like a cell phone conversation skews reaction time distributions, such that the late responses are particularly delayed (Ratcliff & Strayer 2014). Research by Brown et al. (2001) found that these delayed brake reactions significantly elevate both the likelihood and the severity of motor vehicle crashes.

SITUATIONAL AWARENESS

A motorist's situational awareness is shaped by the SPIDER-related processes illustrated in **Figure 1**. The degradation of situational awareness is influenced by both the mental and the temporal demands of the secondary task being performed, with more significant impacts resulting from longer and more demanding tasks.

In **Figure 1**, the bidirectional arrows connecting scanning, predicting, and identifying to situational awareness correspond to Endsley's (1995) three levels of situational awareness. Level 1 involves the perception of elements in the current situation, level 2 pertains to the comprehension of that situation, and level 3 relates to the prediction of its future status. These bidirectional links signify that scanning, predicting, and identifying continually update the driver's mental model, which in turn informs the allocation of attention to each of these processes (Kahneman 1973).

The fidelity of a motorist's situational awareness of the driving situation directly influences their decision-making and the speed of their responses. Endsley (1995) found that even minor lapses in situational awareness can lead to poor performance. Additionally,

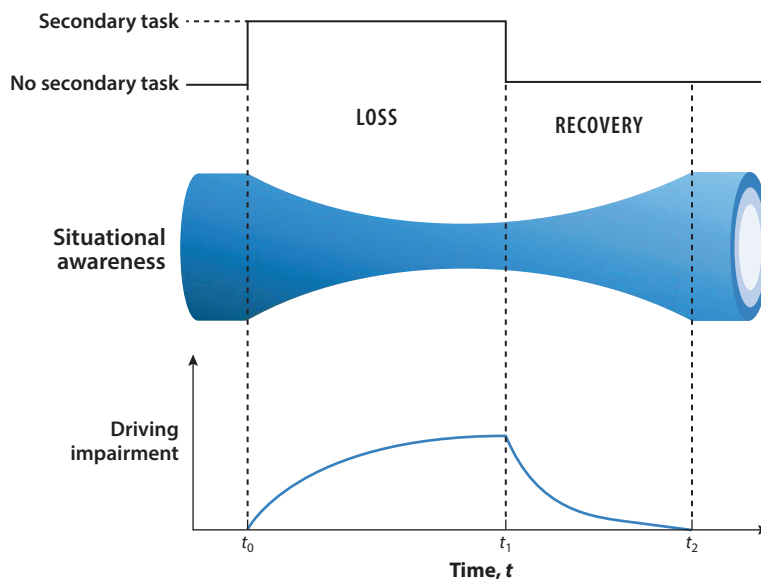


Figure 2

The dynamic fluctuations in attention occur when a motorist engages in an attention-demanding secondary task. The diameter of the cylinder at any given moment represents the level of situational awareness, with a larger diameter indicating a higher level of awareness. A cross section of the cylinder at any moment reflects the situational awareness depicted in the SPIDER (scanning, predicting, identification, decision-making, and executing a response) model (see **Figure 1**). In this figure, situational awareness is at its peak at time t_0 , when the motorist begins to engage in the secondary task. At time t_1 , the secondary task is completed. Between t_0 and t_1 , situational awareness systematically declines, as illustrated by the decreasing diameter of the cylinder, which correlates with a progressive degradation in driving performance. From t_1 to t_2 , situational awareness gradually recovers, as indicated by the increasing diameter of the cylinder.

Fisher & Strayer (2014) discovered that a mere 5% decrease in the likelihood of successfully completing any of the SPIDER-related processes could double the relative risk of a crash. Thus, the accuracy of a motorist's situational awareness is crucial for safe driving.

Figure 2 illustrates the fluctuations in attention to the driving task throughout a multitasking episode. This figure is grounded in information theory (e.g., Hick 1952, Hyman 1953, Shannon 1948, Wickens 1984), which describes the interplay between bandwidth, processing speed, and the information loss resulting from capacity limitations. The diameter of the cylinder in the figure reflects the moment-to-moment level of attention devoted to driving; a larger diameter signifies a higher allocation of attention to the driving task. In the period leading up to t_0 , the motorist is engaged in the single task of driving, with a high level of attention directed toward it. At t_0 , the driver initiates an attention-demanding secondary task (i.e., begins multitasking), resulting in a diversion of focus away from the critical processes necessary for safe driving. Between t_0 and t_1 , driving impairments due to multitasking increase. At t_1 , the motorist ceases the secondary task, allowing for attention to be redirected back to the primary driving task. However, the impairments to driving performance gradually diminish between t_1 and t_2 . Recent research by Strayer et al. (2022b) found that the distraction resulting from engaging in a secondary task while driving can persist for over 30 s after the secondary task has ended. This lingering distraction represents a hidden cost of multitasking. Therefore, the impact of performing a functionally independent secondary task on a driver's situational awareness extends across the interval from t_0 to t_2 .

CONCLUSIONS

Motor vehicle crashes are the leading cause of accidental injury deaths in the United States (NHTSA 2015). According to the National Motor Vehicle Crash Causation Survey, 94% of crashes on US roadways are attributed to human error (Singh 2015). Recognition errors, including driver inattention, distraction, and inadequate surveillance, account for 41% of crashes. Decision errors, such as driving too fast for conditions, making illegal maneuvers, and misjudging gaps or other vehicles' speeds, contribute to 33% of crashes. Performance errors, like lateral or longitudinal control, account for 11%, while nonperformance errors, such as fatigue or falling asleep at the wheel, make up 7%. Other driver errors account for the remaining 8%.

Most of these crashes are preventable and stem from failures in SPIDER-related processes.² As illustrated in **Figure 1**, a driver's allocation of attention to these fundamental tasks is one of the most significant factors influencing roadway safety. When drivers fail to dedicate adequate attention to driving, they are more prone to making mistakes that lead to crashes, injuries, and fatalities.

Driving provides an excellent opportunity to examine visual attention in an everyday context, where multitasking can, and often does, have adverse consequences. The task of driving provides an important context in which to demonstrate if, how, and when the basic science on attention and multitasking scales from the laboratory to the real world. The so-called scaling problem refers to the difficulty in applying the findings from controlled laboratory studies to larger, more complex real-world situations (e.g., Nature Editorial 2019). Here we show that the controlled laboratory studies of attention and multitasking in driving scale well: Even small decreases in object detection and identification and slight delays in responding can cascade into an increase in crashes and fatalities on the roadway.

SUMMARY POINTS

1. Driving is a complex multisensory task, with visual attention playing a critical role in a motorist's situational awareness. Multitasking degrades this awareness, leading to missed and delayed reactions to roadway hazards.
2. Engaging in secondary tasks, such as cell phone use, diverts attention from the driving task. Sources of distractions include visual (looking away), manual (hands off the wheel), and cognitive (mental focus).
3. Safe driving relies on visual scanning, hazard prediction, identification of objects in the line of sight, good decision-making, and timely execution of a response (SPIDER). Distractions degrade these processes, leading to diminished situational awareness and an increased risk of crashes.
4. Multitasking can cause inattention blindness, where drivers fail to see objects in plain sight. Engaging in secondary tasks also slows responses to unexpected road hazards. The effect persists even after the distraction ends, delaying the recovery of situational awareness.
5. Distractions impair the ability to judge safe gaps in traffic, make lane changes, and respond quickly in emergencies, increasing accident risks, especially at intersections.

²A preventable crash is one in which the driver did not take all reasonable measures to avoid the crash.

6. Methods like eye tracking, visual occlusion tests, and electroencephalography (EEG) help assess how drivers allocate visual attention and the impact of distractions on their awareness.

FUTURE ISSUES

1. How does frequent multitasking affect long-term driving behavior and skill retention? Are habitual multitasking drivers less attentive, even when they are not actively distracted?
2. Do some drivers handle distractions better than other drivers? How do factors like age, experience, cognitive abilities, and personality traits influence distraction-related crash risk?
3. How do drivers adapt to new in-vehicle technologies and driver assistance systems? Do automated systems that reduce cognitive load inadvertently increase reliance on secondary tasks (e.g., texting, browsing)?
4. Most research focuses on cell phone use, but how do newer distractions (e.g., voice assistants, infotainment systems, augmented reality dashboards) affect driving performance?
5. How does the complexity of a driving environment (urban versus rural, high-traffic versus low-traffic) interact with different types of distractions? Is there a threshold beyond which distractions become overwhelmingly dangerous, regardless of a driver's skill level?
6. Can brain-based measures (e.g., EEG, functional magnetic resonance imaging) be used to predict distraction-related accidents in real-world settings? Can biometrics such as heart rate variability or eye tracking effectively predict when a driver is too distracted to react safely?
7. Does partial automation (e.g., adaptive cruise control, lane-keeping assist) affect driver attention and response times? What are the potential risks of overreliance on automation, and how do drivers manage the transition between automated and manual driving?

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

LITERATURE CITED

- Adrian ED, Matthews BH. 1934. The Berger rhythm: potential changes from the occipital lobes in man. *Brain* 57(4):355–85
- Amini RE, Al Haddad C, Batabyal D, Gkena I, De Vos B, et al. 2023. Driver distraction and in-vehicle interventions: a driving simulator study on visual attention and driving performance. *Accid. Anal. Prev.* 191:107195
- Anderson DE, Ghate DA, Rizzo M. 2021. Vision, attention, and driving. In *Handbook of Clinical Neurology*, Vol. 178: *Neurology of Vision and Visual Disorders*, ed. JJS Barton, A Leff. Elsevier

- Atchley P, Tran AV, Salehinejad MA. 2017. Constructing a publicly available distracted driving database and research tool. *Accid. Anal. Prev.* 99(Part A):306–11
- Atwood J, Guo F, Fitch G, Dingus TA. 2018. The driver-level crash risk associated with daily cellphone use and cellphone use while driving. *Accid. Anal. Prev.* 119:149–54
- Baker SP, Chen L-H, Li G. 2007. *Nationwide review of graduated licensing*. Tech. Rep., AAA Found. Traffic Saf. <https://aaafoundation.org/wp-content/uploads/2018/02/NationwideReviewOfGDLReport.pdf>
- Benedetto S, Pedrotti M, Minin L, Baccino T, Re A, Montanari R. 2011. Driver workload and eye blink duration. *Transp. Res. F Traffic Psychol. Behav.* 14(3):199–208
- Bergen B, Medeiros-Ward N, Wheeler K, Drews F, Strayer DL. 2013. The crosstalk hypothesis: Language interferes with driving because of modality-specific mental simulation. *J. Exp. Psychol. Gen.* 142:119–30
- Berger H. 1929. Über das Elektroenkephalogramm des Menschen. *Archiv. Psychiatr.* 87:527–70
- Biondi FN, Horrey WJ, Donmez B. 2024a. Preface to the special issue on assessment and effectiveness of driver monitoring systems. *Hum. Factors* 66(9):2161–65
- Biondi FN, Jajo N. 2024. On the impact of on-road partially-automated driving on drivers' cognitive workload and attention allocation. *Accid. Anal. Prev.* 200:107537
- Biondi FN, McDonnell A, Cooper J, Strayer DL. 2024b. Using the ISO detection response task to measure the cognitive load of driving four separate vehicles on two distinct highways. *Transp. Res. F Traffic Psychol. Behav.* 102:260–69
- Biondi F, Turrill JM, Coleman JR, Cooper JM, Strayer DL. 2015. Cognitive distraction impairs driver's anticipatory glances: an on-road study. In *Proceedings of the Eighth International Symposium on Human Factors in Driver Assessment, Training, and Vehicle Design*, ed. DV McGehee, JD Lee, M Rizzo. Public Policy Center, Univ. Iowa
- Braver TS. 2012. The variable nature of cognitive control: a dual mechanisms framework. *Trends Cogn. Sci.* 16:106–13
- Braver TS, Gray JR, Burgess GC. 2008. Explaining the many variables of working memory variation: dual mechanisms of cognitive control. In *Variation in Working Memory*, ed. A Conway, C Jarrod, M Kane, A Miyake, J Towse. Oxford Univ. Press
- Braver TS, Paxton JL, Locke HS, Barch DM. 2009. Flexible neural mechanisms of cognitive control within human prefrontal cortex. *PNAS* 106:7351–56
- Briggs GF, Hole GJ, Turner JAJ. 2017. The impact of attentional set and situation awareness on dual tasking driving performance. *Transp. Res. F Traffic Psychol. Behav.* 57:36–47
- Brooks JR, Passaro AD, Kerick SE, Garcia JO, Franaszczuk PJ, Vettel JM. 2018. Overlapping brain network and alpha power changes suggest visuospatial attention effects on driving performance. *Behav. Neurosci.* 132(1):23–33
- Brown TL, Lee JD, McGehee DV. 2001. Human performance models and rear-end collision avoidance algorithms. *Hum. Factors* 43(3):462–82
- Bruyas M-P, Dumont L. 2013. Sensitivity of detection response task (DRT) to the driving demand and task difficulty. In *Proceedings of the Seventh International Driving Symposium on Human Factors in Driver Assessment, Training, and Vehicle Design*. Public Policy Center, Univ. Iowa
- Caird JK, Simmons SM, Wiley K, Johnson KA, Horrey WJ. 2018. Does talking on a cell phone, with a passenger, or dialing affect driving performance? An updated systematic review and meta-analysis of experimental studies. *Hum. Factors* 60(1):101–33
- Caird JK, Willness CR, Steel P, Scialfa C. 2008. A meta-analysis of the effects of cell phones on driver performance. *Accid. Anal. Prev.* 40:1282–93
- Castro S, Cooper J, Strayer D. 2016. Validating two assessment strategies for visual and cognitive load in a simulated driving task. *Proc. Hum. Factors Ergon. Soc. Annu. Meet.* 60(1):1899–903
- Castro SC, Strayer DL, Matzke D, Heathcote A. 2019. Cognitive workload measurement and modeling under divided attention. *J. Exp. Psychol. Hum. Percept. Perform.* 45:826–29
- Choudhary P, Velaga NR. 2019. Gap acceptance behavior at unsignalized intersections: effects of using a phone and a music player while driving. *Traffic Inj. Prev.* 20(4):372–77
- Cooper JM, Crabtree KW, McDonnell AS, May D, Strayer SC, et al. 2023. Driver behavior while using Level 2 vehicle automation: a hybrid naturalistic study. *Cogn. Res. Princ. Implic.* 8(71):71

- Cooper JM, Medeiros-Ward N, Strayer DL. 2013. The impact of eye movements and cognitive workload on lateral position variability in driving. *Hum. Factors* 55(5):1001–14
- Cooper JM, Strayer DL. 2025. Multitasking induced contextual blindness. *Hum. Factors* 67(3):225–45
- Cooper JM, Vladislavljec I, Medeiros-Ward N, Martin PT, Strayer DL. 2009. An investigation of driving while conversing on a cell phone in simulated highway traffic of varying densities. *Hum. Factors* 51:261–68
- Cooper PJ, Zheng Y. 2002. Turning gap acceptance decision-making: the impact of driver distraction. *J. Saf. Res.* 33:321–35
- Crundall D, Underwood G. 2011. Visual attention while driving: measures of eye movements used in driving research. In *Handbook of Traffic Psychology*, ed. BE Porter. Academic
- D'Addario P, Donmez B. 2019. The effect of cognitive distraction on perception-response time to unexpected abrupt and gradually onset roadway hazards. *Accid. Anal. Prev.* 127:177–85
- Di Campli San Vito P, Shakeri G, Brewster S, Pollick F, et al. 2019. Haptic navigation cues on the steering wheel. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Assoc. Comput. Mach. <https://doi.org/10.1145/3290605.3300440>
- Drews FA, Pasupathi M, Strayer DL. 2008. Passenger and cell phone conversations in simulated driving. *J. Exp. Psychol. Appl.* 14(4):392–400
- Durso F, Rawson K, Giroto S. 2007. Comprehension and situation awareness. In *The Handbook of Applied Cognition*, ed. FT Durso, R Nickerson, ST Dumais, S Lewandowsky, T Perfect. Wiley. 2nd ed.
- Ebadi Y, Fisher DL, Roberts SC. 2019. Impact of cognitive distractions on drivers' hazard anticipation behavior in complex scenarios. *Transp. Res. Rec.* 2673(9):440–51
- Endsley MR. 1995. Towards a theory of situation awareness in dynamic systems. *Hum. Factors* 37(1):32–64
- Endsley MR. 2015. Situation awareness misconceptions and misunderstandings. *J. Cogn. Eng. Decis. Mak.* 9:4–32
- Engström J, Johansson E, Östlund J. 2005. Effects of visual and cognitive load in real and simulated motorway driving. *Transp. Res. F Traffic Psychol. Behav.* 8:97–120
- Evans L. 2004. *Traffic Safety*. Science Serving Society
- Farmer CM, Braver ER, Mitter EL. 1997. Two-vehicle side impact crashes: the relationship of vehicle and crash characteristics to injury severity. *Accid. Anal. Prev.* 29:399–406
- Fisher DL. 2015. How to reduce older driver crashes at intersections by 50%: a proven method and new theory. Paper presented at the Seventh Biennial World Research Congress on the Relationship Between Vision and the Safe Operation of a Motorized Vehicle, Sep. 9
- Fisher DL, Strayer DL. 2014. Modeling situation awareness and crash risk. *Ann. Adv. Automot. Med.* 5:33–39
- Foley JP. 2008. Now you see it, now you don't: visual occlusion as a surrogate distraction measurement technique. In *Driver Distraction: Theory, Effects, and Mitigation*, ed. MA Regan, JD Lee, K Young. CRC Press
- Foxe JJ, Snyder AC. 2011. The role of alpha-band brain oscillations as a sensory suppression mechanism during selective attention. *Front. Psychol.* 2:154
- Foy HJ, Runham P, Chapman P. 2016. Prefrontal cortex activation and young driver behaviour: a fNIRS study. *PLOS ONE* 11(5):e0156512
- Gaffary Y, Lécuyer A. 2018. The use of haptic and tactile information in the car to improve driving safety: a review of current technologies. *Frontiers ICT* 5:5
- Geitner C, Biondi F, Skrypchuk L, Jennings P, Birrell S. 2019. The comparison of auditory, tactile, and multimodal warnings for the effective communication of unexpected events during an automated driving scenario. *Transp. Res. F Traffic Psychol. Behav.* 65:23–33
- Gershon P, Sita KR, Zhu C, Ehsani JP, Klauer SG, et al. 2019. Distracted driving, visual inattention, and crash risk among teenage drivers. *Am. J. Prev. Med.* 56(4):494–500
- Gilroy S, Jones E, Glavin M. 2019. Overcoming occlusion in the automotive environment—a review. *IEEE Trans. Intell. Trans. Syst.* 22(1):23–35
- Goh SYM, Lye SW, Sou KL, Luo C, Xu H. 2021. Effects of increased cognitive load on field of view in multi-task operations involving surveillance. *Hum. Intell. Syst. Integr.* 3:277–85
- Goldman RI, Stern JM, Engel J Jr, Cohen MS. 2002. Simultaneous EEG and fMRI of the alpha rhythm. *Neuroreport* 13(18):2487–92

- Gonçalves RC, Louw TL, Madigan R, Quaresma M, Romano R, Merat N. 2022. The effect of information from dash-based human-machine interfaces on drivers' gaze patterns and lane-change maneuvers after conditionally automated driving. *Accid. Anal. Prev.* 174:106726
- Gonçalves RC, Louw TL, Quaresma M, Madigan R, Merat N. 2020. The effect of motor control requirements on drivers' eye-gaze pattern during automated driving. *Accid. Anal. Prev.* 148:105788
- Grüner M, Ansorge U. 2017. Mobile eye tracking during real-world night driving: a selective review of findings and recommendations for future research. *J. Eye Mov. Res.* 10(2). <https://doi.org/10.16910/jemr.10.2.1>
- Gugerty L. 2011. Situation awareness in driving. In *Handbook for Driving Simulation in Engineering, Medicine, and Psychology*, ed. J Lee, M Rizzo, D Fisher, J Carid. CRC Press
- Haque MM, Oviedo-Trespalacios O, Debnath AK, Washington S. 2016. Gap acceptance behavior of mobile phone-distracted drivers at roundabouts. *Transp. Res. Rec.* 2602:43–51
- Haque MM, Washington S. 2014. A parametric duration model of the reaction times of drivers distracted by mobile phone conversations. *Accid. Anal. Prev.* 62:42–53
- Harbluk JL, Noy Y, Trbovich PL, Eizenman M. 2007. An on-road assessment of cognitive distraction: impacts on drivers' visual behavior and braking performance. *Accid. Anal. Prev.* 39:372–79
- He J, Becic E, Lee Y-C, McCarley JS. 2011. Mind wandering behind the wheel: performance and oculomotor correlates. *Hum. Factors* 53(1):13–21
- Hedlund J, Shults RA, Compton R. 2003. What we know, what we don't know, and what we need to know about graduated driver licensing. *J. Saf. Res.* 34:107–15
- Herslund MB, Jorgensen JO. 2002. Looked-but-failed-to-see-errors in traffic. *Accid. Anal. Prev.* 35:885–91
- Hick WE. 1952. On the rate of gain of information. *Q. J. Exp. Psychol.* 4(1):11–26
- Ho C, Spence C. 2005. Assessing the effectiveness of various auditory cues in capturing a driver's visual attention. *J. Exp. Psychol. Appl.* 11(3):157–74
- Horrey WJ, Wickens CD, Consalus KP. 2006. Modeling drivers' visual attention allocation while interacting with in-vehicle technologies. *J. Exp. Psychol. Appl.* 12(2):67–78
- Howard ZL, Evans NJ, Innes RJ, Brown SD, Eidels A. 2020. How is multi-tasking different from increased difficulty? *Psychon. Bull. Rev.* 27:937–51
- Hutchinson BT, Pammer K, Bandara K. 2020. tACS stimulation at alpha frequency selectively induces inattentional blindness. *Brain Topogr.* 33(3):317–26
- Hutchinson BT, Pammer K, Jack B. 2021. Pre-stimulus alpha predicts inattentional blindness. *Conscious. Cogn.* 87:103034
- Hyman R. 1953. Stimulus information as a determinant of reaction time. *J. Exp. Psychol.* 45(3):188–96
- Ishigami Y, Klein RM. 2009. Is a hands-free phone safer than a handheld phone? *J. Saf. Res.* 40(2):157–64
- ISO (International Organization for Standardization). 2010. *Road vehicles - Ergonomic aspects of transport information and control systems - Simulated lane change test to assess in-vehicle secondary task demand*. Draft International Standard 26022:2010, ISO TC22/SC39/WG8. <https://www.iso.org/standard/43361.html>
- ISO (International Organization for Standardization). 2016. *Road Vehicles -Transport information and control systems - Detection Response Task (DRT) for assessing selective attention in driving*. Draft International Standard 17488:2016, ISO TC 22/SC39/WG8. <https://www.iso.org/standard/59887.html>
- Jones PR, Ungewiss J, Eichinger P, Wörner M, Crabb DP, Schiefer U. 2022. Contrast sensitivity and night driving in older people: quantifying the relationship between visual acuity, contrast sensitivity, and hazard detection distance in a night-time driving simulator. *Front. Hum. Neurosci.* 16:914459
- Ju U, Chuang LL, Wallraven C. 2020. Acoustic cues increase situational awareness in accident situations: a VR car-driving study. *IEEE Trans. Intell. Transp. Syst.* 23(4):3281–91
- Kahneman D. 1973. *Attention and Effort*. Prentice-Hall
- Kapitaniak B, Walczak M, Kosobudzki M, Józwiak Z, Bortkiewicz A. 2015. Application of eye-tracking in drivers testing: a review of research. *Int. J. Occup. Med. Environ. Health* 28(6):941–54
- Kass SJ, Cole KS, Stanny CJ. 2007. Effects of distraction and experience on situation awareness and simulated driving. *Transp. Res. F Traffic Psychol. Behav.* 10:321–29
- Koch I, Poljac E, Müller H, Kiesel A. 2018. Cognitive structure, flexibility, and plasticity in human multitasking—an integrative review of dual-task and task-switching research. *Psychol. Bull.* 144:557–83

- Kujala T, Kircher K, Ahlström C. 2023. A review of occlusion as a tool to assess attentional demand in driving. *Hum. Factors* 65(5):792–808
- Kujala T, Sarkar A. 2024. Spare visual capacity and driver inattention in dynamic car following scenarios. *Transp. Res. F Traffic Psychol. Behav.* 104:506–21
- Land MF, Lee DN. 1994. Where we look when we steer. *Nature* 369(6483):742–44
- Large DR, Crundall E, Burnett G, Harvey C, Konstantopoulos P. 2016. Driving without wings: the effect of different digital mirror locations on the visual behaviour, performance and opinions of drivers. *Appl. Ergon.* 55:138–48
- Leynes PA, Flynn J, Mok BA. 2018. Event-related potential measures of smartphone distraction. *Cyberpsychol. Behav. Soc. Netw.* 21(4):248–53
- Lohani M, Payne BR, Strayer DL. 2019. A review of psychophysiological measures to assess cognitive states in real-world driving. *Front. Hum. Neurosci.* 13:57
- Louw T, Merat N. 2017. Are you in the loop? Using gaze dispersion to understand driver visual attention during vehicle automation. *Transp. Res. C Emerg. Technol.* 76:35–50
- Mack A, Rock I. 1998. *Inattentional Blindness*. MIT Press
- Mackenzie AK, Harris JM. 2017. A link between attentional function, effective eye movements, and driving ability. *J. Exp. Psychol. Hum. Percept. Perform.* 43(2):381–94
- Matias J, Belletier C, Izaute M, Lutz M, Silvert L. 2022. The role of perceptual and cognitive load on inattention blindness: a systematic review and three meta-analyses. *Q. J. Exp. Psychol.* 75(10):1844–75
- Mazzae EN, Satterfield KS, Baldwin GHS, Skuce IA, Andrella A. 2023. Driver eye glance behavior and performance with camera-based visibility systems versus mirrors. *Traffic Inj. Prev.* 24(1):S94–99
- McCarley JS, Vais M, Pringle H, Kramer AF, Irwin DE, Strayer DL. 2001. Conversation disrupts visual scanning of traffic scenes. Paper presented at the 9th International Conference on Vision in Vehicles
- McCarley JS, Vais M, Pringle H, Kramer AF, Irwin DE, Strayer DL. 2004. Conversation disrupts scanning and change detection in complex visual scenes. *Hum. Factors* 46(3):424–36
- McDonnell AS, Crabtree KW, Cooper JM, Strayer DL. 2023. This is your brain on autopilot 2.0: the influence of practice on driver workload and engagement during on-road, partially automated driving. *Hum. Factors* 66(8):2025–40
- McDonnell AS, Imberger K, Poulter C, Cooper JM. 2021a. The power and sensitivity of four core driver workload measures for benchmarking the distraction potential of new driver vehicle interfaces. *Transp. Res. F Traffic Psychol. Behav.* 83:99–117
- McDonnell AS, Simmons TG, Erickson GG, Lohani M, Cooper JM, Strayer DL. 2021b. This is your brain on autopilot: neural indices of driver workload and engagement during partial vehicle automation. *Hum. Factors* 65(7):1435–50
- McEvoy SP, Stevenson MR, McCartt AT, Woodward M, Haworth C, et al. 2005. Role of mobile phones in motor unit crashes resulting in hospital attendance: a case-crossover study. *Br. Med. J.* 331:428–33
- Medeiros-Ward N, Cooper JM, Strayer DL. 2014. Hierarchical control and driving. *J. Exp. Psychol. Gen.* 143:953–58
- Mole CD, Wilkie RM. 2017. Looking forward to safer HGVs: the impact of mirrors on driver reaction times. *Accid. Anal. Prev.* 107:173–85
- Moslemi N, Soryani M, Azmi R. 2021. Computer vision-based recognition of driver distraction: a review. *Concurr. Comput. Pract. Exp.* 33(24):e6475
- National Center for Statistics and Analysis. 2021. *Distracted Driving 2019*. Rep. DOT HS 813 111, National Highway Traffic Safety Administration
- Nature Editorial. 2019. From the lab to real-world use. *Nat. Sustain.* 2:989
- Navarro J, Hernout E, Osiurak F, Reynaud E. 2020. On the nature of eye-hand coordination in natural steering behavior. *PLOS ONE* 15(11):e0242818
- Navon D, Gopher D. 1979. On the economy of the human-processing system. *Psychol. Rev.* 86:214–55
- Navon D, Miller J. 1987. Role of outcome conflict in dual-task interference. *J. Exp. Psychol. Hum. Percept. Perform.* 13:435–48
- NHTSA (National Highway Traffic Safety Administration). 2015. *Motor vehicle traffic crashes as a leading cause of death in the United States, 2010 and 2011* (DOT HS 812 203). Rep., NHTSA. <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812203>

- NHTSA (National Highway Traffic Safety Administration). 2023. Fatality and Injury Reporting System Tool (FIRST), Version 5.6. <https://www.nhtsa.gov/document/nhtsas-fatality-and-injury-reporting-system-tool-first>
- Norman DA, Bobrow DG. 1975. On data-limited and resource-limited processes. *Cogn. Psychol.* 7:44–64
- NSC (National Safety Council). 2018. *Vulnerable road users*. Policy statement, NSC. <https://www.nsc.org/getattachment/d5babe6-582d-4e66-804f-8d06f9b021a4/t-vulnerable-road-users-147?srsid=AfmBOoqjQ78ThByODZgWkKKBdnPipf4bN8VzksDb8ChyopOQBS45cnc>
- NSC (National Safety Council). 2024. *Understanding distracted driving*. Rep., NSC. <https://www.nsc.org/faforms/ddam-understanding-driver-distraction-white-paper>
- Pashler H. 1994a. Dual-task interference in simple tasks: data and theory. *Psychol. Bull.* 116:220–44
- Pashler H. 1994b. Graded capacity-sharing in dual-task interference? *J. Exp. Psychol. Hum. Percept. Perform.* 20:330–42
- Peaylo C, Hilla Y, Sauseng P. 2021. Cause or consequence? Alpha oscillations in visuospatial attention. *Trends Neurosci.* 44(9):705–13
- Posner MI, Boies SJ. 1971. Components of attention. *Psychol. Rev.* 78:391–408
- Posner MI, Keele SW. 1967. Decay of visual information from a single letter. *Science* 158:137–39
- Pizzighello S, Bressan P. 2008. Auditory attention causes inattention blindness. *Perception* 37:859–66
- Ranney TA, Baldwin GH, Smith LA, Mazzae EN, Pierce RS. 2014. *Detection response task (DRT) evaluation for driver distraction measurement application*. Rep. DOT HS 812 077, National Highway Traffic Safety Administration. <https://www.nhtsa.gov/sites/nhtsa.gov/files/documents/812077.pdf>
- Ratcliff R, Strayer DL. 2014. Modeling simulated driving with a one-boundary diffusion model. *Psychol. Bull. Rev.* 21:577–89
- Readinger WO, Chatziastros A, Cunningham DW, Bülthoff HH, Cutting JE. 2002. Gaze-eccentricity effects on road position and steering. *J. Exp. Psychol. Appl.* 8:247–58
- Recarte MA, Nunes LM. 2000. Effects of verbal and spatial-imagery tasks on eye fixations while driving. *J. Exp. Psychol. Appl.* 6:31–43
- Redelmeier DA, Tibshirani RJ. 1997. Association between cellular-telephone calls and motor unit collisions. *N. Engl. J. Med.* 336:453–58
- Regan MA, Hallett C, Gordon CP. 2011. Driver distraction and driver inattention: definition, relationship and taxonomy. *Accid. Anal. Prev.* 43:1771–81
- Reimer B. 2009. Impact of cognitive task complexity on drivers' visual tunneling. *Transp. Res. Rec.* 2138(1):13–19
- Reimer B, Mehler B, Wang Y, Coughlin JF. 2012. A field study on the impact of variations in short-term memory demands on drivers' visual attention and driving performance across three age groups. *Hum. Factors* 54(3):454–68
- Rogers SD, Kadar EE, Costall A. 2005. Gaze patterns in the visual control of straight-road driving and braking as a function of speed and expertise. *Ecol. Psychol.* 17(1):19–38
- Rolison JJ, Regev S, Moutari S, Feeney A. 2018. What are the factors that contribute to road accidents? An assessment of law enforcement views, ordinary drivers' opinions, and road accident records. *Accid. Anal. Prev.* 115:11–24
- Romei V, Brodbeck V, Michel C, Amedi A, Pascual-Leone A, Thut G. 2008. Spontaneous fluctuations in posterior α -band EEG activity reflect variability in excitability of human visual areas. *Cereb. Cortex* 18(9):2010–18
- Salvucci DD. 2019. Modeling driver distraction from cognitive tasks. In *Proceedings of the Twenty-Fourth Annual Conference of the Cognitive Science Society*, ed. WD Gray, CD Schunn. Routledge
- Saryazdi R, Bak K, Campos JL. 2019. Inattention blindness during driving in younger and older adults. *Front. Psychol.* 10:880
- Sauseng P, Klimesch W, Stadler W, Schabus M, Doppelmayr M, et al. 2005. A shift of visual spatial attention is selectively associated with human EEG alpha activity. *Eur. J. Neurosci.* 22(11):2917–26
- Senders JW, Kristofferson AB, Levison WH, Dietrich CW, Ward JL. 1967. The attentional demand of automobile driving. *Highway Res. Rec.* 195:15–32
- Shamini PB, Vinodhini M, Keerthana B, Lakshna S, Meenatchi KR. 2022. Driver drowsiness detection based on monitoring of eye blink rate. In *2022 4th International Conference on Smart Systems and Inventive Technology (ICSSIT)*. IEEE. <https://ieeexplore.ieee.org/document/9716304>

- Shannon CE. 1948. A mathematical theory of communication. *Bell Syst. Tech. J.* 27:379–423
- Sifrit KJ, Stutts J, Martell C, Staplin L. 2011. *Intersection crashes among drivers in their 60s, 70s and 80s*. Rep. DOT HS 911 495, National Highway Traffic Safety Administration. <https://www.nhtsa.gov/sites/nhtsa.gov/files/811495.pdf>
- Singh S. 2015. *Critical reasons for crashes investigated in the National Motor Vehicle Crash Causation Survey*. Rep. DOT HS 812 115, National Highway Traffic Safety Administration. <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812506>
- Simons DJ. 2000. Attention capture and inattention blindness. *Trends Cogn. Sci.* 4:147–55
- Simons DJ, Chabris CF. 1999. Gorillas in our midst: sustained inattention blindness for dynamic events. *Perception* 28(9):1059–74
- Simons-Morton B, Guo G, Klauer SG, Ehsani JP, Pradhan AK. 2014. Keep your eyes on the road: young driver crash risk increases according to duration of distraction. *J. Adolesc. Health* 54(5 Suppl.):S61–67
- Sirevaag EJ, Kramer AF, Coles MGH, Donchin E. 1989. Resource reciprocity: an event-related brain potentials analysis. *Acta Psychol.* 70:77–97
- Staplin L, Lyles RW. 1991. Age differences in motion perception and specific traffic maneuver problems. *Transp. Res. Rec.* 1325:23–33
- Strayer DL. 2015. Is the technology in your car driving you to distraction? *Policy Insights Behav. Brain Sci.* 2:156–65
- Strayer DL, Biondi F, Cooper JM. 2017. Dynamic workload fluctuations in driver/non-driver conversational dyads. In *Proceedings of the Ninth International Symposium on Human Factors in Driver Assessment, Training, and Vehicle Design*, ed. DV McGehee, JD Lee, M Rizzo. Public Policy Center, Univ. Iowa
- Strayer DL, Castro SC, McDonnell AS. 2022a. The multitasking motorist. In *The Handbook of Human Multitasking*, ed. A Kiesel, L Johannsen, H Müller, I Hoch. Springer
- Strayer DL, Castro SC, Turrill J, Cooper JM. 2022b. The persistence of distraction: the hidden costs of intermittent multitasking. *J. Exp. Psychol. Appl.* 28:262–82
- Strayer DL, Cooper JS. 2015. Driven to distraction. *Hum. Factors* 57:1343–47
- Strayer DL, Cooper JM, Drews FA. 2004. What do drivers fail to see when conversing on a cell phone? *Proc. Hum. Factors Ergon. Soc. Annu. Meet.* 48(19):2213–17
- Strayer DL, Cooper JM, Goethe RM, McCarty MM, Getty D, Biondi F. 2019. Assessing the visual and cognitive demands of using in-vehicle information systems. *Cogn. Res. Princ. Implic.* 4:18
- Strayer DL, Cooper JM, McCarty MM, Getty DJ, Wheatley CL, et al. 2018. *Visual and cognitive demands of using Apple's CarPlay, Google's Android Auto and five different OEM infotainment systems*. Tech. Rep., AAA Found. Traffic Saf. <https://aaafoundation.org/wp-content/uploads/2018/06/AAA-Phase-6-CarPlay-Android-Auto-FINAL.pdf>
- Strayer DL, Drews FA. 2007a. Attention. In *Handbook of Applied Cognition*, ed. F Durso, R Nickerson, S Dumais, S Lewandowsky, T Perfect. Wiley. 2nd ed.
- Strayer DL, Drews FA. 2007b. Cell-phone induced inattention blindness. *Curr. Dir. Psychol. Sci.* 16:128–31
- Strayer DL, Drews FA, Johnston WA. 2003. Cell phone induced failures of visual attention during simulated driving. *J. Exp. Psychol. Appl.* 9:23–52
- Strayer DL, Fisher DL. 2016. SPIDER: a model of driver distraction and situation awareness. *Hum. Factors* 58(1):5–12
- Strayer DL, Johnston WA. 2001. Driven to distraction: dual-task studies of simulated driving and conversing on a cellular phone. *Psychol. Sci.* 12:462–66
- Strayer DL, Turrill J, Cooper JM, Coleman J, Medeiros-Ward N, Biondi F. 2015. Assessing cognitive distraction in the automobile. *Hum. Factors* 57(8):1300–24
- Thut G, Nietzel A, Brandt SA, Pascual-Leone A. 2006. α -Band electroencephalographic activity over occipital cortex indexes visuospatial attention bias and predicts visual target detection. *J. Neurosci.* 26(37):9494–502
- Tillman G, Strayer DL, Eidels A, Heathcote A. 2017. Modeling cognitive load effects between a passenger and driver. *Atten. Percept. Psychophys.* 79:1759–803
- Toscani M, Marzi T, Righi S, Viggiano MP, Baldassi S. 2010. Alpha waves: a neural signature of visual suppression. *Exp. Brain Res.* 207:213–19
- Tsai Y, Viirre E, Strychacz C, Chase B, Jung T. 2007. Task performance and eye activity: predicting behavior relating to cognitive workload. *Aviation Space Environ. Med.* 5:B176–85

- Van Winsum W. 2018. The effects of cognitive and visual workload on peripheral detection in the detection response task. *Hum. Factors* 60(6):855–69
- Van Winsum W. 2019. Optic flow and tunnel vision in the detection response task. *Hum. Factors* 61(6):992–1003
- Veitch E, Alsos OA. 2023. Gorillas on a boat: inattention blindness during supervisory control of autonomous vessels. *J. Phys. Conf. Ser.* 2618:012010
- Victor TW, Harbluk JL, Engström JA. 2005. Sensitivity of eye-movement measures to in-vehicle task difficulty. *Transp. Res. F Traffic Psychol. Behav.* 8:167–90
- Wang M, Liao Y, Lyckvi SL, Chen F. 2020. How drivers respond to visual versus auditory information in advisory traffic information systems. *Behav. Inform. Technol.* 39(12):1308–19
- Wang P, Chan CY, de La Fortelle A. 2018. A reinforcement learning based approach for automated lane change maneuvers. Preprint, arXiv:1804.07871 [cs.RO]. <https://arxiv.org/abs/1804.07871>
- Wang Y, Reimer B, Dobres J, Mehler B. 2014. The sensitivity of different methodologies for characterizing drivers' gaze concentration under increased cognitive demand. *Transp. Res. F Traffic Psychol. Behav.* 26:227–37
- Watson JM, Miller AE, Lambert A, Strayer DL. 2011. The magical letters P, F, C, and sometimes U: the rise and fall of executive attention with the development of prefrontal cortex. In *Handbook of Lifespan Psychology*, ed. K Fingerman, C Berg, T Antonucci, J Smith. Springer
- Wickens CD. 1980. The structure of attentional resources. In *Attention and Performance*, ed. R Nickerson. Erlbaum
- Wickens CD. 1984. Processing resources in attention. In *Varieties of Attention*, ed. R Parasuraman, R Davies. Academic
- Wickens CD. 2008. Multiple resources and mental workload. *Hum. Factors* 50(3):449–55
- Wilson M, Chattington M, Marple-Horvat DE. 2008. Eye movements drive steering: Reduced eye movement distribution impairs steering and driving performance. *J. Motor Behav.* 40:190–202
- Wolfe B, Dobres J, Rosenholtz R, Reimer B. 2017. More than the useful field: considering peripheral vision in driving. *Appl. Ergon.* 65:316–25
- Wolfe JM, Kosovicheva A, Wolfe B. 2022. Normal blindness: when we look but fail to see. *Trends Cogn. Sci.* 26:809–19
- Wood JM. 2020. Nighttime driving: visual, lighting and visibility challenges. *Ophthalmic Physiol. Optics* 40(2):187–201
- Wright TJ, Samuel S, Borowsky A, Zilberstein S, Fisher DL. 2016. Experienced drivers are quicker to achieve situation awareness than inexperienced drivers in situations of transfer of control within a Level 3 autonomous environment. *Proc. Hum. Factors Ergon. Soc. Annu. Meet.* 60(1):270–73
- Xin X, Wang Y, Liu N, Yang W, Dong H, Liu W. 2021. Research on in-vehicle haptic interactions as crucial resources for driver perceptions. In *HCI International 2021-Late Breaking Papers: HCI Applications in Health, Transport, and Industry*, ed. C Stephanidis, VG Duffy, H Krömker, FF-H Nah, K Siau, et al. Springer
- Yahoodik S, Tahami H, Unverricht J, Yamani Y, Handley H, Thompson D. 2021. Blink rate as a measure of driver workload during simulated driving. *Proc. Hum. Factors Ergon. Soc. Annu. Meet.* 64(1):1825–28
- Zheng H, Hu F, Green P. 2023. Assessing driver distraction: enhancements of the ISO 26022 lane change task to make its difficulty adjustable. SAE Tech. Pap. 2023-01-0791. <https://doi.org/10.4271/2023-01-0791>